

Group Theory

With Okron's Templates

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1. Introduction

¶ Note. 1.1

All counters (Theorem, Definition, Proposition, etc.) are automatically reset at each section heading (level 1).

1.1. Group Initiation

¶ Def. 1.1 (*Group*)

A **group** is an ordered pair $(G, *)$ where G is a set and $*$: $G \times G \rightarrow G$ is a binary operation such that:

1. **Associativity:** $(a * b) * c = a * (b * c)$ for all $a, b, c \in G$.
2. **Identity:** There exists $e \in G$ such that $a * e = e * a = a$ for all $a \in G$.
3. **Inverse:** For each $a \in G$ there exists $a^{\{-1\}} \in G$ with $a * a^{\{-1\}} = a^{\{-1\}} * a = e$.

With the previous definition we are now able to make some observations, as the following:

¶ Prop. 1.1 (*Uniqueness of identity and inverses*)

In a group $(G, *)$ the identity element is unique, and every element has a unique inverse.

And also we can talk about Lemmas.

¶ Lemma. 1.1 (*Powers in a group*)

For any $a \in G$ and $m, n \in \mathbb{Z}$, we have $a^m a^n = a^{m+n}$ and $(a^m)^n = a^{mn}$, where a^k is defined recursively.

Finally it may be useful to write some Affirmations.

¶ Aff. 1.1 (*Left and right cancellation*)

In a group, $ab = ac$ implies $b = c$, and $ba = ca$ implies $b = c$.

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2. Theorems and Proofs

§ **Theorem. 2.1** (*Lagrange's Theorem*)

If G is a finite group and $H \leq G$ a subgroup, then $|H|$ divides $|G|$. In particular, $|G| = [G : H] \cdot |H|$.

§ **Corollary. 2.1** (*Order of an element*)

For any element $g \in G$ of finite order, $|g|$ divides $|G|$.

§ **Proof. 2.1**

Let $n = |g|$ and let $H = \langle g \rangle$ be the cyclic subgroup generated by g . Then $|H| = n$. By Lagrange's theorem n divides $|G|$.

■

§ **Remark. 2.1** The converse of Lagrange's theorem does not hold in general. For example, A_4 has order 12 but has no subgroup of order 6.

3. Examples and Exercises

‡ **Example. 3.1** (*Cyclic group*)

The group $\mathbb{Z}_n$ under addition modulo n is cyclic. Every subgroup of a cyclic group is also cyclic.

‡ **Exercise. 3.1** (*A non-abelian group*)

Show that the symmetric group S_3 is not abelian.

‡ **Proof. 3.1**

S_3 consists of permutations of three elements. Compute $(12)(13) = (132)$ but $(13)(12) = (123)$. Since $(132) \neq (123)$, the group is non-abelian.

■

‡ **Exercise. 3.2** (*Subgroup test*)

Prove that a non-empty subset $H \subseteq G$ is a subgroup iff $ab^{-1} \in H$ whenever $a, b \in H$.

4. Figures, Code and Plots



Figure 1: The best profile picture ever.

If the image is missing, you can still see the figure structure.

¶ **Code. 4.1** (*Python Multiplication*)

```
def multiply(a, b):  
    return a * b  
  
result = multiply(3, 5)  
print(f"3 * 5 = {result}")
```

→ **stdout:**

```
3 * 5 = 15
```

This is a practical way to include code snippets on your writings.